

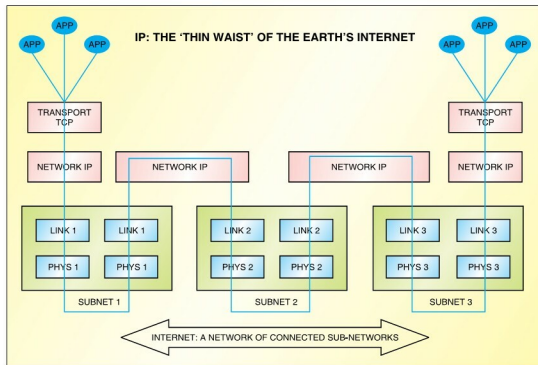


Fig. 2: Working of the Internet

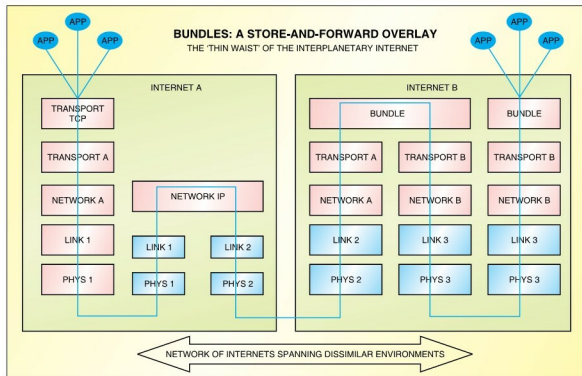


Fig. 3: Working of IPN

tion. Forwarding means sending bundles on demand for non-persistent nodes. Custody transfer is reliable intra-IPN delivery with storage for persistent nodes. In fact, in custodial transfer, a sender can transmit a file to a receiver over a single link and on receipt of the entire file, the receiver notifies the sender that it will take care of any suc-

cessive forward-transmission hop.

IPN nodes

Current IPN nodes are agent, relay, gateway and custody transfer. The agent builds and consumes bundles. Relays are agents that can forward bundles within or between regions. Gateways are relays that do routing

between regions. Custody transfers are orthogonal and optional versus nodes.

Alternative IPN nodes may be non-persistent (NP), persistent (P) and exception (E). The NP node may have no stable storage, and may build and consume bundles, forward or cache bundles in full or parts, participate in time synchronisation and never assume custody. The P node has stable storage, does everything what NP nodes do, accepts custody transfer and notifies prior custodian of custody transfer. The E nodes accept custody always.

Typically, end-to-end delivery from a source (S) to a destination (D) may take place as:

1. 'S' determines the IPN's next hop 'H,' next custodian 'C' and sending time from the IPN router
2. 'S' gets the IPN service time ticket using send time
3. 'S' arranges for acknowledgement from 'C'
4. If 'H' is a 'P' node, 'H' will send a custody transfer notification and 'S' will free its resources
5. If 'H' is an 'NP' node, it will forward to the next hop

IPN nodes face challenges of instability of the backbone network and very long-haul link

(which could introduce long delay and errors). So an interplanetary backbone would be a set of high-capacity, high-availability links between network traffic hubs, like terrestrial network backbones. The difference is that the hubs and nodes in the IPN would in many cases be hundreds of millions of kilometres apart.

Obviously, as all interplanetary